

NEET 2018
Set NN
Solution – Physics

1.

$$U = \frac{1}{2}LI^2$$

$$25 \times 10^{-3} = \frac{1}{2}L(60 \times 10^{-3})^2$$

$$\Rightarrow L = \frac{50 \times 10^{-3}}{3600 \times 10^{-6}}$$

$$= \frac{25 \times 10}{103}$$

$$= \frac{13.89H}{(a)}$$

2.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{q}$$

$$\Rightarrow -\frac{1}{15} = \frac{1}{v} - \frac{1}{40}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{15} + \frac{1}{40}$$

$$\Rightarrow v = \frac{-600}{25}$$

$$v = -24 \text{ cm}$$

Finally

$$\frac{1}{15} = \frac{1}{v^1} - \frac{1}{20}$$

$$\Rightarrow -\frac{1}{15} + \frac{1}{20} = \frac{1}{v^1}$$

$$\Rightarrow v^1 = \frac{300}{5}$$

So image is displaced by 36 cm away from mirror (d)

3.

Direction of propagation of Em wave is same as

$$\vec{E} \times \vec{B}$$

$$\text{So, } \hat{i} = \hat{j} \times (B)$$

So direction of B is along k

Ans (d)

4.

According to question

Ray will retrace its path if $r_2 = 0$

Also,

$$A = r_1 + r_2$$

$$\Rightarrow 30^\circ = r_1$$

By small law,

$$u = \frac{\sin i}{\sin 30^\circ}$$

$$\Rightarrow \sqrt{2} = \frac{\sin i}{\frac{1}{2}}$$

$$\Rightarrow i = 45^\circ$$

Ans (d)

5.

$$V_{BE} = 0$$

$$V_{CE} = 0$$

$$V_b = 0$$

$$I_C = \frac{(20 - 0)}{4 \times 10^3}$$

$$I_C = 5 \times 10^{-3} = 5mA$$

$$V_i = V_{BE} + I_R R_B$$

$$20 = I_B \times 500 \times 10^3$$

$$I_B \frac{20}{500 \times 10^3} = 40 \mu A$$

$$\beta = \frac{I_C}{I_b} = \frac{25 \times 10^{-3}}{40 \times 10^{-6}} = 125$$

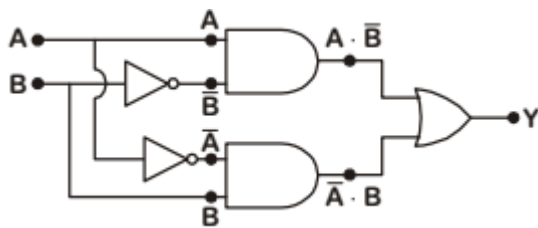
Ans(1)

6. Due to heating, number of electron-hole pairs will increase, so overall resistance of diode will change.

Due to which forward biasing and reversed biasing both are charged.

Ans(1)

7.



$$Y = (A\bar{B} + \bar{A}B)$$

Ans(4)

8. We know,

$$\lambda_{\max} T = \text{constant (wien's law)}$$

So,

$$\lambda_{\max 1} T_1 = \lambda_{\max 2} T_2$$

$$\Rightarrow \lambda_0 T = \frac{3\lambda_0}{4} T'$$

$$\Rightarrow T' = \frac{4}{3} T$$

$$\text{So, } \frac{P_2}{P_1} = \left(\frac{T'}{T}\right)^4 = \left(\frac{4}{3}\right)^4 = \frac{256}{81}$$

Ans(3)

9. Accd to ques

$$Y = \frac{F \times L}{A \Delta L}$$

Since vol. of both wired is same So,

$$A \times L = 3A \times L'$$

$$= L' = \frac{\Delta}{3},$$

$$Y = \frac{F \times L}{A \Delta L} = \frac{F' \times \frac{L}{3}}{3A \times \Delta L}$$

$$F' = 9F$$

Ans (2)

10.

$$\Delta u = \Delta q - \Delta w$$

$$\Delta q = 54 \times 4.2$$

$$= 226.8 \text{ J}$$

$$\Delta w = p \Delta v$$

$$v_i = \frac{0.1}{1} = 0.1 \text{ cc}$$

$$\Rightarrow \Delta v = 167 \text{ cc}$$

$$= 167 \times 10^{-6} \text{ m}^3$$

$$\Delta w = 16.91 \text{ J}$$

$$\therefore \Delta w = P \Delta v$$

$$\Rightarrow \Delta u = 226.8 - 16.91$$

$$= 209.89 \text{ J Ans (4)}$$

11.

$$\text{Rate of prod of heat} = F \times u$$

$$= 6\pi n r v \times v$$

$$= 6\pi n r \left[\frac{2}{9} \frac{r^2 g}{n} (p_b - p_i) \right]$$

$$\text{So, Rate of prod of heat} \propto r^5 \text{ Ans (3)}$$

12.

Accd to Ensein's Eq,

$$h \times 2v_0 = w_0 + \frac{1}{2} m v_1^2 \dots (1)$$

$$\& h \times 5v_0 = w_0 + \frac{1}{2} m v_1^2 \dots (2)$$

$$\text{Since } w_0 = h v_0$$

$$\text{From (1), } h v_0 = \frac{1}{2} m v_1^2$$

$$\& \text{From (2) } h \times 4v_0 = \frac{1}{2} m v_2^2$$

$$\Rightarrow 4 = \frac{v_2^2}{v_1^2}$$

$$\Rightarrow \frac{v_2}{v_1} = 2$$

Ans (2)

13. After 450 nukes disintegrates

$$\text{No of nukes remaining} = 600 - 450$$

$$= 150$$

$$\& 150 = \frac{600}{4}$$

$$\text{So, } N = N_0 \times \left(\frac{1}{2}\right)^1$$

$$\Rightarrow 150 = 600 \times \left(\frac{1}{2}\right)^1$$

$$\Rightarrow \text{No of half lives} = 2$$

$$\begin{aligned} \text{So five taken} &= 2 \times 10 \\ &= 20 \text{ min} \quad \text{Ans (2)} \end{aligned}$$

14.

$$\begin{aligned} &\longrightarrow v_0 \hat{i} \\ &\xrightarrow{E = -E_0 \hat{i}} \end{aligned}$$

So, electron being a negative charge So, force exp, by it along the direction of its velocity so electron after 't' sec is,

$$V^1 = V_0 + \left(\frac{Ee}{m}\right)t$$

$$\text{So, } \lambda_o = \frac{h}{mv_0}$$

$$\lambda_o = \frac{h}{m\left(v_1 + \frac{eE}{m}t\right)} = \frac{h}{mv_0 + (eft)}$$

$$\Rightarrow \lambda^1 = \frac{h / mvo}{1 + \frac{eET}{mv_0}} = \frac{v_0}{1 + \frac{eEt}{mv_0}}$$

Ans (2)

15.

Since, Total energy = - (Kinetic energy)

$$\text{So, } \frac{k.e}{t.e} \quad 1 : - 1$$

Ans (4)

16.

This is a closed pipe

$$\text{Atv, } 320 = \frac{v}{4\{20 + e\}} = \frac{3v}{4[73 + e]}$$

E = end correction

$$\Rightarrow 73 + e = 60 + 3e$$

$$\Rightarrow e = \frac{13}{2} \text{ cm} = 6.5 \text{ cm}$$

$$\text{So, } 320 = \frac{v}{4 \times \left(\frac{26.5}{100} \right)}$$

$$\Rightarrow v = 320 \times 4 \times \frac{26.5}{100} = 339 \text{ m/s}$$

17.

$$F = \frac{\sigma^2}{2\epsilon_0} A$$

So it is independent of distance between plates

Ans (2)

18. For SHM,

$$Q = + W^2 Y$$

$$\Rightarrow 20 = W^2 \times 5$$

$$\Rightarrow W = 2 = \frac{2T_1}{T}$$

$$\Rightarrow T = T_1$$

Ans (4)

19.

$$\text{Time of fall} = t = \sqrt{\frac{2h}{a}}$$

$$\text{Where } a = \frac{eE}{m}$$

$$\Rightarrow t = \sqrt{\frac{2h}{eE}} \times n$$

$$\Rightarrow \frac{t_e}{t_p} = \sqrt{\frac{m_e}{m_p}} \approx \frac{1}{42}$$

$$\Rightarrow t_e < t_p \quad \text{Ans (2)}$$

20.

When planet revolves around star,

$Mvr = \text{constant}$

$$\text{So, } v \propto \frac{1}{r}$$

Since r_c is max of V_c is Min

So, V_A is Max

$$\text{So, } K_A > K_B > K_C$$

Ans (4)

21.

For body rolling,

$$K_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}mv^2 + \frac{1}{2} \times \frac{2}{5}MR^2 \times \frac{V^2}{R^2}$$

$$K_{\text{total}} = \frac{7}{10}mv^2$$

$$\text{So, } \frac{K_t}{K_{\text{total}}} = \frac{\frac{1}{2}mv^2}{\frac{7}{10}mv^2} = \frac{5}{7}$$

Ans (4)

22.

$$g = \frac{GM}{R^2}, \quad M = \text{Mass of earth}$$

Of G becomes 10 times, g becomes 10 times

$$\text{Time period of pendulum } T = 2\pi\sqrt{\frac{L}{g}}$$

So T decreases.

Ans (3)

23.

Since radius is increasing while sphere is rotating freely so, Torque = 0

$$\text{Hence angular} = \frac{\text{Angular Momentum}}{\text{Constant}}$$

24.

For rod to remain at rest,

$$mg \sin 30^\circ = F \cos 30^\circ$$

$$\Rightarrow mg \times \frac{1}{2} = BIl \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow I = \left(\frac{m}{l} \right) g \times \frac{1}{\sqrt{3}} \times \frac{1}{B}$$

$$= 0.5 \times 10 \times \frac{1}{\sqrt{3}} \times 4$$

$$I = 11.32 \text{ A}$$

Ans (1)

25.

Impedence of circuit

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\omega = 314$$

$$\text{So, } \omega L = 314 \times \frac{20}{1000} = 6.28 \text{ OHM}$$

$$\frac{1}{\omega C} = \frac{1}{314 \times 100} \times 10^6 = 31.84 \text{ ohm}$$

$$\Rightarrow Z = \sqrt{(50)^2 + (31.84 - 6.28)^2}$$

$$X = 56 \text{ ohm}$$

Power Loss=

$$= V_{\text{rms}} \times I_{\text{rms}} \times \frac{R}{Z}$$

$$= \frac{10}{\sqrt{2}} \times \frac{10}{\sqrt{2} \times 56} \times \frac{50}{56}$$

$$= 1.13 \text{ w}$$

Ans (1)

26.

Ans (1)

Because when electromagnet is switched in the magnetic field associated with rod increases and leads to generation of induced electric field.

27.

$$R = \frac{\text{Voltage Sensitivity (V / div)}}{\text{Current Sensitivity (A / div)}}$$

$$= \frac{1}{\frac{20}{1}} = 250 \text{ ohm}$$

$$\frac{5000}{1}$$

Ans (3)

28.

Since graph shows,

Vol $\times$ temp

So pressure = constant

$$\Delta Q = n c_p \Delta T = n \times \frac{5}{2} R \Delta T.$$

$$\{ \text{For mono atomic gas, } C_p = \frac{5}{2} R \}$$

$$C_v = \frac{3}{2} R$$

$$W = nR \Delta T$$

$$\text{So, } \frac{W}{\Delta Q} = \frac{2}{5} \text{ Ans (2)}$$

29.

According to questions,

$$\frac{L}{2L_0} = 3 \frac{L}{4L_c}$$

$$\Rightarrow L_0 = \frac{4L_c}{3 \times 2} = \frac{40}{3} = 13.3 \text{ cm}$$

Ans (2)

30.

$$m = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{273}{373} = \frac{100}{373} = 26.8\%$$

31.

For escaping out from earth,

$V_{rms} = v_{escape}$

$$\Rightarrow \sqrt{\frac{3KT}{M}} = 11.2 \times 10^3$$

$$\Rightarrow \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times T}{22.76 \times 10^{-26}}} = 11200$$

$$\Rightarrow \frac{3T \times 1000}{2} = 11200 \times 11200$$

$$\Rightarrow T \approx 8.36 \times 10^4 \text{ K}$$

32.

Since reflected & refracted rays are mutually perp

So, $\mu = \tan(i)$

So, Ans (4)

33.

$$\text{Ang Fringe width} = \frac{\beta}{D} = \frac{\lambda}{d}$$

$$\Rightarrow 0.2 = \frac{\lambda}{2}$$

$$\& 0.21 = \frac{\lambda}{d^1}$$

$$\Rightarrow \frac{0.2}{0.21} = \frac{d^1}{2} \Rightarrow d^1 = 1.9 \text{ mm}$$

Ans (4)

34.

For telescope

$$M = \frac{f_o}{f_e}$$

$$\& \text{Resolving power} = \frac{D}{1.22\lambda}$$

So, Ans (3)

35.

For body to complete vertical circle

$$V_A = \sqrt{5Rg} = \sqrt{\frac{5}{2}gD}$$

$$\text{So, } \sqrt{2gH} = \sqrt{\frac{5}{2}gD}$$

$$\Rightarrow 2gH = \frac{5}{2}gD$$

$$\Rightarrow H = \frac{5}{4}D$$

Ans (1)

36.

Accd to work energy theorem,

$$W = k_f - k_i$$

Since $k_f = 0$

$$W = -k_i$$

$$K_A = \frac{1}{2} \times I \omega^2 = \frac{1}{2} \times \frac{2}{5} MR^2 \omega^2$$

$$K_B = \frac{1}{2} \times \frac{1}{2} MR^2 \omega^2$$

$$K_C = \frac{1}{2} MR^2 \omega^2$$

So, $W_C > W_B > W_A$

Ans (2)

37.

Ans (1)

μ is dimensionless

38.

By conservation of linear momentum,

$$mv + 4m \times 0 = m \times 0 + 4m \times v'$$

$$\Rightarrow v' = \frac{v}{4}$$

$$\text{So, } e = \left| \frac{\frac{v}{4} - 0}{\frac{v}{4} - 0} \right| = \frac{1}{4} = 0.25$$

Ans (4)

39.

$$R = 47000 \, \Omega \pm 4700 \, \Omega$$

So colour code is yellow violet orange silver

($\because$ Tolerance = 10%)

40.

According to question

$$I = \frac{E}{nR + R} = \frac{E}{R(n+1)}$$

$$\& 10I = \frac{E}{\frac{R}{n} + R} = \frac{E}{R\left[1 + \frac{1}{n}\right]}$$

$$\Rightarrow 10 = \frac{1}{\left(1 + \frac{1}{n}\right)} = \frac{n+1}{1 + \frac{1}{n}} = n$$

$$n = 10$$

Ans (2)

41.

When battery is short circuited

$$I = \frac{ne}{nr} = \frac{e}{r}$$

It is independent of 'n'

Ans (2)

42.

$$\text{Acceleration, } a = \frac{qE}{m}$$

$$v = u + at$$

$$\Rightarrow 6 = 0 + \frac{qE}{m} \times 1$$

$$\Rightarrow \frac{qE}{m} = 6 = q \dots\dots\dots(1)$$

From $t = 1$ to $t = 3$, the body retards,

$$V_f = 6 = \frac{qE}{m} \times 2 = -6 \text{ m/s}$$

$$u = 0 \rightarrow v = 6 \text{ m/s} \rightarrow v = 0$$

$$t = 0 \quad s_1 \quad t = 1 \quad s_2 \quad t = 2$$

$$t = 3 \leftarrow s_3$$

$$CV = 6 \text{ m/s}$$

$$S_1 = \frac{1}{2} \times a \times 1^2 = \frac{1}{2} a \left[s = ut + \frac{1}{2} at^2 \right]$$

$$S_2 = \frac{0^2 - 6^2}{2(-a)} = \frac{18}{a} \left[S = \frac{v^2 - u^2}{2a} \right]$$

$$S_3 = \frac{1}{2} \times 9 \times 1^2 = \frac{1}{2} a \left[S = Ut + \frac{1}{2} at^2 \right]$$

$$\text{Avg Speed} = \frac{\frac{1}{2}a + \frac{18}{a} + \frac{1}{2}a}{3}$$

Putting $a = 6$

$$\Rightarrow \text{Avg speed} = 3 \text{ m/s}$$

$$\text{Avg velocity} = \frac{\frac{1}{2}a + \frac{18}{a} + \frac{1}{2}a}{3}$$

$$= 1 \text{ m/s}$$

Ans (4)

43.

When wedge accelerate s , block, experiences pseudo force in opp direction,

$\Rightarrow$ For block, to remain at rest,

$$mg \sin \theta = ma \cos \theta$$

$$\Rightarrow \boxed{a = g \tan \theta}$$

44.

L. count = 0.001 cm zero error = - 0.004 cm

MSR = 5mm

$$\text{Circular scale Reading} = 25 \times 0.001 \text{ cm} \\ = 0.025 \text{ cm}$$

$$\text{Diameter} = 0.5 + (0.5 + 25 \times 0.001) \times 0.001$$

$$\text{So, } d = 0.529 \text{ cm}$$

Ans (1)

45.

$$\hat{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$$

$$\hat{r} = 2\hat{j} - \hat{k}$$

$$\text{Torque} = \hat{r} \times \hat{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & k \\ 0 & 2 & -1 \\ 4 & 5 & -6 \end{vmatrix}$$

$$\hat{i}(-7) - \hat{j}(4) + k(-8)$$

Ans (1)

